Tushar Mishra

+91-8009751879 | mishrat859@gmail.com | LinkedIn | GitHub

Education

Vellore Institute of Technology — Bhopal, Madhya Pradesh
B.Tech in Computer Science and Engineering — 2023 – 2027 (Expected)
CGPA: 8.56

Class XII – CBSE — 2020 – 2021
Divine Public School, Maharajganj, UP — Percentage: 93.6%

Class X – CBSE — 2018 – 2019
St. Joseph's School, Maharajganj, UP — Percentage: 85.6%

Technical Skills

Programming Languages: C++
Frontend Development: HTML, CSS, Tailwind CSS, JavaScript, React.js, Context API, Streamlit
Backend Development: Node.js, Express.js, RESTful APIs, Mongoose
Databases and Vector Search: MongoDB, SQL, FAISS
AI/LLM Technologies: Python, RAG, LangChain, Hugging Face, Sentence Transformers, Groq API, Llama
Authentication & Security: JWT, bcrypt, OTP Email Verification
API Integrations & Tools: Google Gemini API, Nodemailer, Puppeteer, Multer, PyPDFLoader

Personal Projects

MediBot – AI Medical Chatbot using RAG *2026*

Technologies: Python, Streamlit, LangChain, FAISS, Hugging Face, Sentence Transformers, Groq API, Llama

- Built a Streamlit medical chatbot using Retrieval-Augmented Generation (RAG) to answer questions from a medical encyclopedia PDF using retrieved source text.
- Created a PDF ingestion pipeline with LangChain's PyPDFLoader and recursive chunking (500-character chunks, 50-character overlap); generated Sentence Transformers embeddings and indexed them in FAISS for top-3 semantic retrieval.
- Integrated LangChain's RetrievalQA chain with a Groq-hosted Llama model and a context-focused prompt to generate answers from relevant medical passages.
- Displayed retrieved source documents with answers, maintained chat history using Streamlit session state, and cached the FAISS vector store between interactions.

InterviewAI – AI-Powered Interview Preparation Platform *2025*

Technologies: React.js, Node.js, Express.js, MongoDB, Google Gemini API, JWT, Puppeteer

- Developed an AI-powered interview preparation platform that analyzes uploaded PDF resumes, candidate profiles, and job descriptions to generate personalized interview preparation reports.
- Enforced structured AI output with Zod-derived JSON schemas, producing a 0–100 job-match score, technical and behavioral questions with interviewer intent and model answers, severity-rated skill gaps, and a day-wise preparation roadmap stored in MongoDB.
- Implemented PDF text extraction using PDF-Parse and MongoDB-based report storage, enabling users to retrieve previous interview reports through secure REST APIs.
- Built an AI-powered, job-tailored resume generation feature that produces ATS-friendly HTML resumes and converts them into downloadable PDF documents using Puppeteer.
- Implemented user authentication using JWT, bcrypt password hashing, protected API routes, and token blacklisting to manage secure login and logout functionality.

Achievements

- **LeetCode:** Solved 300+ Data Structures and Algorithms problems and earned the 200 Days Coding Badge.
- **GeeksforGeeks:** Solved 100+ Data Structures and Algorithms problems.

Additional Information

- **Languages:** Hindi, English.
- **Hobbies:** Listening to music, Cricket, Cooking.